

MODEL QUESTION PAPER				
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY				
SECOND SEMESTER B. TECH DEGREE EXAMINATION, MONTH AND YEAR				
Course Code : PCITT205				
Course Name: DISCRETE MATHEMATICAL STRUCTURES				
Max. Marks: 60			Duration: 2 hours 30 minutes	
PART A				
		Answer all questions. Each question carries 3 marks	CO	Marks
1		Using truth table, verify the logical equivalence $[p \rightarrow (q \vee r)] \Leftrightarrow [\neg r \rightarrow (p \rightarrow q)]$	CO1	(3)
2		Define the terms: Converse, Inverse and Contrapositive.	CO1	(3)
3		What is the pigeonhole principle? Explain. If you select any five numbers from 1 to 8 then prove that at least two of them will add up to 9.	CO2	(3)
4		In how many ways can the letters of the word 'SOCIOLOGICAL' be arranged such that all the vowels are together?	CO2	(3)
5		Define poset with an example.	CO3	(3)
6		Let $S = \{0,1,2,3,4,\dots\}$. Is $(S, +)$ a monoid?	CO4	(3)
7		Find the generating function for the sequence 1,1,1,...,1,0,0,0,... where the first n+1 terms are 1	CO5	(3)
8		A bank pays 6% interest on savings, compounding the interest yearly. Write the recurrence relation for this and solve it to find how much will a deposit of Rs 1000 be worth after 12 years.	CO5	(3)
PART B				
Answer any one full question from each module. Each question carries 9 marks				
Module 1				
9	a)	Without using truth tables, show that $(p \vee q) \wedge \neg(\neg p \wedge q) \Leftrightarrow p$	CO1	(5)
	b)	Let $p(x)$, $q(x)$ and $r(x)$ be the following open statements $p(x): x^2 - 7x + 10 = 0$, $q(x): x^2 - 2x - 3 = 0$, $r(x): x < 0$ Determine the truth or falsity of the following statements, where the universe is all integers. If a statement is false, provide a counter example or explanation. (i) $\forall x[p(x) \rightarrow \neg r(x)]$ (ii) $\forall x[q(x) \rightarrow r(x)]$ (iii) $\exists x[q(x) \rightarrow r(x)]$ (iv) $\exists x[p(x) \rightarrow r(x)]$	CO1	(4)
10	a)	Establish the validity of the following argument	CO1	(5)

		$ \begin{array}{c} p \rightarrow (q \rightarrow r) \\ pv \neg s \\ q \\ \hline s \rightarrow r \end{array} $		
	b)	Express the negation of the statement using quantifier “if the teacher is absent, then some students do not keep quiet.”	CO1	(4)
		Module 2		
11	a)	Determine the number of integers n , $1 \leq n \leq 2000$ that are not divisible by 2,3 or 5.	CO2	(5)
	b)	Determine the number of integer solutions of $x_1 + x_2 + x_3 + x_4 = 32$, such that (i) $x_i \geq 0$, $1 \leq i \leq 4$ (ii) $x_i > 0$, $1 \leq i \leq 4$	CO2	(4)
12	a)	In how many ways can 10 identical dimes be distributed among five children if (a) there are no restrictions (b) each child gets at least one dime (c) the oldest child gets at least 2 dimes.	CO2	(5)
	b)	Determine the coefficient of $w^3x^2yz^2$ in $(2w - x + 3y - 2z)^8$	CO2	(4)
		Module 3		
13	a)	Show that set Q^+ of all positive rational numbers forms an abelian group under the operation $*$ defined by $a * b = \frac{ab}{2}$, $a, b \in Q^+$	CO4	(5)
	b)	Find the complement of each element in D_{42} .	CO3	(4)
14	a)	Consider the set $A = \{a, b, c\}$. Show that $Q(A)$, the set of all proper subsets of A is a partially ordered set under the relation $\subseteq$, the set inclusion. Draw the Hasse diagram for the poset $(Q(A), \subseteq)$.	CO3	(5)
	b)	For $A = \{1,2,3\}$ and $B = \{w, x, y, z\}$. Give an example for a relation from A to B which is not a function.	CO3	(4)
		Module 4		
15	a)	Solve $a_{n+1} - a_n = 2n + 3$, $n \geq 0$, $a_0 = 1$.	CO5	(5)
	b)	Determine the sequence generated by the exponential generating function $f(x) = \frac{1}{1-x}$	CO5	(4)
16	a)	Solve the recurrence relation $2a_n = 7a_{n-1} - 3a_{n-2}$, $a_0 = 2$, $a_1 = 5$	CO5	(5)
	b)	Solve the recurrence relation $a_{n+1} = 2a_n + 1$, $n \geq 0$, $a_0 = 0$.	CO5	(4)